

YUNXI LAI

Physician-Scientist | Research Associate

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EDUCATION

- Jul 2023–Jun 2026 **DOCTOR OF CLINICAL MEDICINE, Ophthalmology**
Zhongshan Ophthalmic Center, Sun Yat-sen University, China
Thesis: *Transformation from Clinical Data to Scientific Evidence Enabled by Large Language Models in Ophthalmology*
- Sep 2020–Jun 2023 **MASTER OF MEDICINE, Ophthalmology**
Zhongshan Ophthalmic Center, Sun Yat-sen University, China
Thesis: *Analyses of Routine Preoperative Blood Testing in Ocular Patients*
- Sep 2015–Jun 2020 **BACHELOR OF MEDICINE, Clinical Medicine**
Guangdong Medical University, China

RESEARCH EXPERIENCE

Clinical AI and Large Language Models | Jul 2023–Jun 2026

Development, Validation and Implementation of an Ophthalmic Large Language Model

- **Co-developed an ophthalmic large language model and evaluation benchmarks** for assessing question-answering performance; co-authored the resulting article in *JAMA Ophthalmology* 2024.
- **Conceived its clinical application as a pre-consultation tool**, designed an agent to actively elicit and document medical histories, and integrated it into existing outpatient workflows and hospital systems.
- **Led clinical validation through a randomized controlled trial**, including protocol and statistical analysis plan development, ethics and trial registration, Python-based data processing, statistical analysis in R and trial operations; co-first-authored the article in *npj Digital Medicine*.
- **Coordinated multidisciplinary collaboration** across grant applications, intellectual property and public outreach, contributing to app deployment serving over 1.7 million people across four countries and to the project's selection as a Chinese digital health case study at the 2025 World Health Assembly.

Epidemiology | Sep 2021–Jun 2023

Routine Preoperative Blood Testing in Ophthalmic Surgery

- **Independently conducted a retrospective study** evaluating the necessity of routine preoperative blood testing and the applicability of *Choosing Wisely* recommendations to ophthalmic surgery in China.
- **Analyzed and interpreted data from 102,542 patients**, resulting in a first-author publication in the *British Journal of Ophthalmology* 2024.

Cohort Studies | Aug 2023–Mar 2026

Pediatric Cohort for Ocular Monitoring in Systemic Disease and Medication Use

- **Co-led a cross-institutional ophthalmology–pediatrics cohort study** to monitor ocular changes in children with systemic diseases receiving glucocorticoids or hydroxychloroquine; responsibilities included ethics approval, participant recruitment, ophthalmic examinations and longitudinal data management.
- **Co-established and managed a prospective cohort of more than 600 children**, with an average of seven follow-up visits per participant.
- **Designed an LLM-assisted extraction method with active data-quality control** to address inefficient longitudinal data extraction, converting discharge summaries into structured research variables; internally evaluated the method using retrospective inpatient records.

Cohort Studies | Nov 2023–Mar 2025

Community-Based Ophthalmic Cohorts

- Explained test results, offered medical advice and collaborated with local healthcare providers.

TECHNICAL & RESEARCH SKILLS

- **Computational and Statistical:** Python, R, SPSS; data processing, statistical modeling, reproducible analysis; AI-assisted coding and prototyping.
- **Human-Subjects Research:** Designed and conducted human-subject studies, including randomized controlled trials, prospective cohorts and retrospective analyses, from protocol development and registration to statistical analysis planning, data collection and analysis.
- **Clinical and Visual Assessment:** Comprehensive ophthalmic examination and imaging, including slit-lamp biomicroscopy, ophthalmoscopy, optical coherence tomography and scanning laser ophthalmoscopy; experience across emergency, inpatient and outpatient ophthalmic settings.
- **Project Management and Interdisciplinary Collaboration:** Coordinated timelines and deliverables across clinical, technical and industry teams; integrated expertise from different disciplines to advance project development and implementation.
- **Research Presentation:** Drafted, revised and published two first-author manuscripts in peer-reviewed journals; gave and received constructive scientific feedback.
- **Languages:** Chinese (native), English (fluent).

TEACHING & MENTORING

Teaching Assistant | Undergraduate Ophthalmology | Aug 2024 | Sun Yat-sen University

- Assisted in undergraduate ophthalmology teaching, including clinical case discussion, ophthalmic examination, and interpretation of common ocular findings.
- Supported students in linking core ophthalmic knowledge with clinical reasoning and patient assessment.

Instructor | Training for Junior Ophthalmology Residents | Aug 2024

- Prepared teaching materials and provided hands-on instruction in essential ophthalmic clinical skills and procedures.
- Guided junior residents in clinical examination techniques, documentation and preparation for ophthalmology rotations.
- Demonstrated clinical workflows and supported trainees in developing systematic approaches to patient assessment.
- Mentored graduates in clinical research, literature review, data organization and scientific presentation.

SELECTED OUTREACH

Community Engagement and Public Eye Health | 2020–2025

- Contributed to community eye-health screening initiatives and public-facing communication for the ophthalmic LLM project, including content development and media promotion.
- Led visual identity and art direction for the official launch of the ophthalmic LLM and contributed to video planning, scientific content development and media communication for public dissemination.

PUBLICATIONS

Peer-reviewed publications: 8 | Under Review: 1 | First/co-first author: 3

1. Luo, M. J.*, **Lai, Y.***, Wang, W.*, Bi, S., Jin, L., Chen, J., Wu, X., Lin, Z., Wang, J., Pang, J., Dongye, M., Wu, Y., Liu, Z., & Lin, H. Standardized Pre-consultation by a Large Language Model Agent vs Ophthalmology Residents: A Randomized Clinical Trial. *npj Digital Medicine*. (accepted 30 August 2026). *Co-first authors.
2. **Lai, Y.**, Zeng, W., Liao, J., Yu, Y., Liu, X., & Wu, K. (2024). Retrospective Analyses of Routine Preoperative Blood Testing in a Tertiary Eye Hospital: Could Choosing Wisely Work in China? *British Journal of Ophthalmology*, 108(6), 897–902.
3. Pang, J.*, Luo, M. J.*, **Lai, Y.***, Yun, D.*, Cui, T.*, Chen, W., Wu, L., Jin, L., Wu, X., Bi, S., Chen, J., & Lin, H. Augmenting Ophthalmic Hotline Services via a PatientGuide Agent: A Large-Scale, Pragmatic Implementation Study with a Sequential-Cohort Design. *The Lancet Digital Health* (under review). *Co-first authors.

4. Luo, M. J., Pang, J., Bi, S., **Lai, Y.**, Zhao, J., Shang, Y., Cui, T., Yang, Y., Lin, Z., Zhao, L., Wu, X., Lin, D., Chen, J., & Lin, H. (2024). Development and Evaluation of a Retrieval-Augmented Large Language Model Framework for Ophthalmology. *JAMA Ophthalmology*, 142(9), 798–805.
5. Luo, M. J., Bi, S., Pang, J., Liu, L., Tsui, C. K., **Lai, Y.**, Chen, W., Yang, Y., Xu, K., Zhao, L., Jin, L., Lin, D., Wu, X., Chen, J., Chen, R., Liu, Z., Zou, Y., Yang, Y., Li, Y., & Lin, H. (2025). A Large Language Model Digital Patient System Enhances Ophthalmology History Taking Skills. *npj Digital Medicine*, 8(1), 502.
6. Liu, L., Hong, J., Wu, Y., Liu, S., Wang, K., Li, M., Zhao, L., Liu, Z., Li, L., Cui, T., Tsui, C. K., Xu, F., Hu, W., Yun, D., Chen, X., Shang, Y., Bi, S., Wei, X., **Lai, Y.**, Lin, D., Fu, Z., Deng, Y., Cai, K., Xie, Y., Cao, Z., Wang, D., Zhang, X., Dongye, M., Lin, H., & Wu, X. (2024). Digital ray: Enhancing Cataractous Fundus Images Using Style Transfer Generative Adversarial Networks to Improve Retinopathy Detection. *British Journal of Ophthalmology*, 108(10), 1423–1429.
7. Shang, Y., Lin, Y., Li, R., Shang, Y., Li, M., Zhao, L., Jin, L., Xu, A., Liu, D., Wu, Q., Luo, M., Pang, J., Bi, S., He, Y., Xu, M., Chen, X., Cao, Z., Yu, S., Zhao, J., **Lai, Y.**, Chen, W., & Lin, H. (2025). The Effectiveness of Large Language Models in Medical AI Research for Physicians: A Randomized Controlled Trial. *Cell Reports Medicine*, 6(12), 102469.
8. Li, Y., Cui, T., Lin, Z., Zhao, L., **Lai, Y.**, Yun, D., Lin, Z., Li, M., Huang, Y., Chen, S., Wang, D., Qiu, Y., Xu, Y., Yang, Y., Lin, D., Jiang, X., & Lin, H. (2026). Risk Factors and Critical Thresholds for Glucocorticoid-Induced Ocular Adverse Reactions in Children: Implications for Muscle and Neuromuscular Disease Management. *European Journal of Translational Myology*, 36(2).
9. Yang, Y., Han, Y., Wang, R., Ling, S., Lyu, J., Li, C., Zhao, L., Wu, Q., Wen, J., Cui, T., Li, Y., **Lai, Y.**, Zeng, D., Chen, X., Zhang, Y., Ye, L., Yang, W., Gao, X., Hu, J., Gong, C., Zou, J., Meng, J., Li, X., Liang, X., Huang, D., Yu-Wai-Man, P., Xu, F., & Lin, H. (2026). Smartphone-Based 3D Ocular Reconstruction for Automated Quantitative Blepharoptosis Evaluation. *JAMA Ophthalmology*. Published online September 24, 2026.

CONFERENCE PRESENTATION

Lai, Y., Luo, M. J., Yu, Y., Pang, J., Bi, S., Lin, H. *Evaluating GPT-4o in ophthalmic consultation and documentation: Bridging knowledge with clinical practice*. Poster presented at: Association for Research in Vision and Ophthalmology Annual Meeting; May 2025; Salt Lake City, UT, USA.

LICENSURE & CERTIFICATION

Physician License: China.

National Standardized Residency Training Certificate: Ophthalmology.